

SUMMER AI WORKSHOP

Tool Ecosystem

Paul Wayland

Business + Technology @

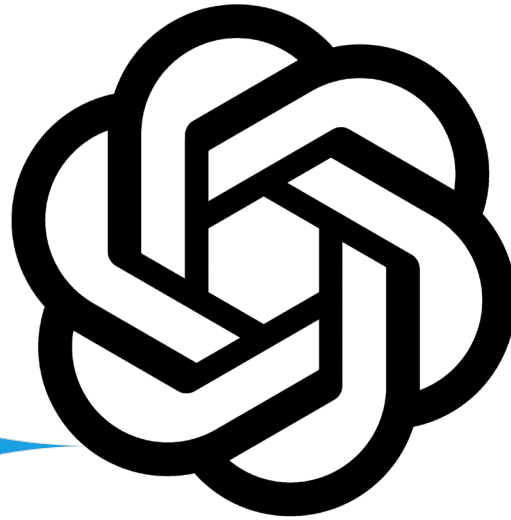




which AI is best
which tool fits which job

Chatbots

TOOL MAP



TOOL MAP

Chatbots

brainstorming summarizing

explaining

tutoring

planning

drafting

role-playing

coding help

revising

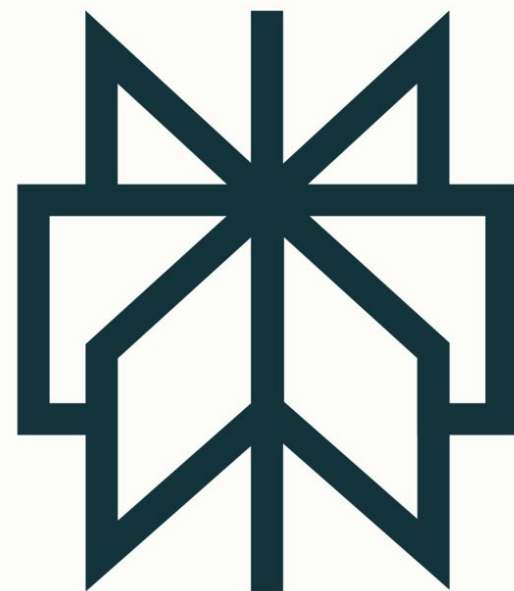
TOOL MAP

Chatbots

Weakness: sound **authoritative** without
being grounded

TOOL MAP

Search-answer tools



TOOL MAP

Search-answer tools

Weakness: citations are **NOT** *magic*

Coding copilots

TOOL MAP



TOOL MAP

Coding copilots

tests

scaffolding

errors

refactoring

file-by-file

debugging

implementation

TOOL MAP

Coding copilots

Weakness: generate working-*looking* code

Office copilots

TOOL MAP



TOOL MAP

Office copilots

spreadsheets

email threads

calendars

documents

organizational data

meetings

Office copilots

TOOL MAP

Weakness: depend on organization **data** and **permissions**

TOOL MAP

Multimodal tools

spreadsheet

screenshot

photo

diagram

document

A TOOL CAN BE

- fluent but wrong
- cited but weak
- correct but poorly explained
- useful but unsafe for private data
- powerful but unavailable because of account limits
- great for brainstorming but bad for current research

WRITING – chatbot

outline

revise

change tone

improve clarity

make examples

create drafts

generate rubrics

check an argument

DON'T submit unexamined
prose

treat AI as a collaborator,
not a ghostwriter

RESEARCH – search-answer tools

library databases
official sources
Google Scholar
institutional pages
government sources
primary documents

general chatbot for orientation

for facts, citations, dates, laws,
policies, and statistics, **use
source-grounded tools and
verification**

CODING – coding copilots

ask
generate
run
inspect
test
revise

general chatbot can explain
concepts and sketch code

tools connected to files and
terminals are better for **real**
projects

NEVER trust code only because
it looks professional

STUDYING – chatbots

explain at different levels

generate practice quizzes

role-play Socratic questioning

create flashcards

summarize notes

make study plans.

ground in course materials

whenever possible

STUDYING – chatbots

Use only these notes unless you clearly
label outside knowledge.

Quiz me on the material.

Do not give the answer until I try first.

BRAINSTORMING – chatbots

ask for many options, strange options, practical options,
and bad ideas

separate generation from evaluation

first produce possibilities, then judge them

use AI to widen the field before you narrow it

PRIVACY

depends on:

- tool
- account type
- settings
- institution contract
- connected apps
- data controls

PRIVACY

OpenAI's ChatGPT privacy pages describe user data controls, temporary chats, and memory controls

PRIVACY

Google's Gemini privacy notice says Gemini Apps may process prompts, uploaded content, files, screens, recordings, connected app data, and location information depending on use/settings

PRIVACY

Anthropic also documents data-use and
privacy controls for Claude

PRIVACY

DON'T paste private, graded, medical, legal, financial, employment, FERPA-protected, or institution-sensitive data into a consumer AI tool UNLESS

you know the policy allows it

CITATIONS

- Does the source actually say what the AI claims?
- Is the source credible?
- Is it current?
- Is it primary or secondary?
- Did the AI quote accurately?
- Is the cited source about the same country, institution, law, or time period?

Account limits are a *live* classroom **risk**

may hit message caps, verification issues,
blocked signups, unavailable tools, missing
browser permissions, or network friction

Account limits are a live classroom **risk**

always have fallbacks: screenshots of tool outputs, allow any available chatbot, have a demo account, make the assignment about comparison

FINAL THOUGHTS

Tool ecosystems are not just technical.
They are also economic, institutional, and practical.

The “best” tool is the one that fits the task, data sensitivity, access, and required level of truth.